

Building and deploying a React application

Building and deploying a React application involves several steps, depending on where you want to deploy it (e.g., **GitHub Pages**, **Netlify**, **Vercel**, **Firebase**, or a **custom server**). Below is a general step-by-step guide that covers **building** and **deploying** a React app.

Step 1: Build the React App

Create a React app (if you haven't already):

```
npx create-react-app my-app
```

```
cd my-app
```

Develop your app by editing the files in the `src` directory.

Build the production-ready app

```
npm run build
```

This creates a `build/` folder with optimized production files.

Step 2: Deploy Options

Install GitHub Pages package: **`npm install --save gh-pages`**

Add the following lines to your `package.json`:

```
"homepage": "https://<your-username>.github.io/<repo-name>",
"scripts": {
  "predeploy": "npm run build",
  "deploy": "gh-pages -d build"
}
```

Initialize a git repo and push to GitHub:

```
git init
```

```
git remote add origin https://github.com/<your-username>/<repo-name>.git
```

```
git add .
```

```
git commit -m "Initial commit"
```

```
git push -u origin main
```

Deploy: **`npm run deploy`**

Option 2: Deploy to Netlify

Step 1: Build your React app

Run this in your project folder:

```
npm run build
```

This creates a `build/` folder with the production-ready app.

Step 2: Go to Netlify website

- Open netlify.com and log in or sign up.
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Step 3: Upload your site manually

1. On your Netlify dashboard, click “**Add new site**” → “**Deploy manually**” or “**Deploy site without Git**”.
 2. Drag and drop the entire `build/` folder into the upload area.
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Step 4: Wait for the upload and deployment

Netlify will process the files and deploy your app instantly.

Step 5: Get your site URL

Once done, Netlify provides a live URL where your React app is accessible.

Optional: Change site name or add a custom domain

- You can rename your site or set up a custom domain from the **Site settings**.

